DEPARTMENT OF ADSE

Integrated IoT for Predictive Applications

CIA-3

TOPIC: Smart Factory Energy and Equipment Utilization System – Business planning with prediction

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1. Introduction

Modern manufacturing facilities face growing challenges in managing energy consumption and equipment utilization efficiently. Idle machines waste electricity, increase operational costs, and reduce overall factory productivity. An IoT-based monitoring system offers a real-time, data-driven solution to detect machine states and optimize energy usage.

This report documents the design, hardware simulation, and implementation of an IoT-Based Factory Energy and Equipment Utilization System using the TinkerCAD online simulation platform and validated on real hardware using an ESP32 microcontroller, OLED display, and DHT sensor.

1.1 Objectives

- Simulate real-time energy monitoring using an analog sensor (potentiometer)
- Detect machine ACTIVE and IDLE states based on energy threshold logic
- Display live Power % and Status on a 16x2 LCD screen
- Provide visual alerts using Green (Active) and Red (Idle) LEDs
- Log data to Serial Monitor to simulate cloud data transmission
- Validate simulation results on real ESP32 hardware with OLED display and DHT sensor

1.2 Scope

The scope of this project covers all four project milestones:

1. Problem Analysis & System Design
2. Hardware Development & Simulation
3. Cloud Integration Logic & Efficiency Analysis
4. Final Reporting with Findings and Recommendations

2. System Design & Architecture

The system is designed as a layered IoT architecture consisting of a sensing layer, processing layer, display/alert layer, and a data logging layer extensible to the cloud.

2.1 System Architecture

Layer	Component	Function
Sensing Layer	Potentiometer (A0)	Simulates current/energy sensor (0–1023 analog value)
Sensing Layer	DHT Sensor (Real HW)	Measures temperature & humidity for environment monitoring
Processing Layer	Arduino Uno R3 / ESP32	Reads sensor data, applies threshold logic, controls outputs
Display Layer	16x2 LCD / OLED	Shows Power %, Temperature, Humidity, Machine Status
Alert Layer	Green LED (Pin 8)	Lights when machine is ACTIVE (energy > threshold)
Alert Layer	Red LED (Pin 9)	Lights when machine is IDLE (energy <= threshold)
Data Layer	Serial Monitor / WiFi	Logs sensor values — simulates or transmits to cloud

2.2 Threshold Logic

The system uses threshold comparison to determine machine state:

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If sensorValue > 300  →  Status: ACTIVE  →  Green LED ON
If sensorValue <= 300  →  Status: IDLE     →  Red LED ON
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3. Components & Specifications

	Component	Qty	Specifications	Uses
1	Arduino Uno R3	1	ATmega328P, 14 digital I/O, 6 analog	Main microcontroller (simulation)
2	ESP32 DevKit	1	Xtensa LX6, WiFi+BT, 38 pins	Cloud-connected MCU (real hardware)
3	16x2 LCD Display	1	HD44780, 5V, 4-bit mode	Real-time data display (simulation)
4	OLED Display (0.96")	1	SSD1306, I2C, 128x64px	Real-time display (real hardware)
5	DHT Sensor	1	DHT11/DHT22, Temp & Humidity	Environment monitoring (real hardware)
6	Potentiometer	1	10kΩ, 3-terminal	Simulates analog energy sensor
7	Green LED	1	5mm, 2.2V forward voltage	Machine ACTIVE indicator
8	Red LED	1	5mm, 1.8V forward voltage	Machine IDLE indicator
9	Resistors (220Ω)	2	1/4W carbon film	LED current limiting
10	Breadboard	1	830 tie-point full size	Component & wiring base

4. Circuit Connections — Step by Step

Step 1 — Power Rails

- Arduino 5V pin → Breadboard positive (+) rail
- Arduino GND pin → Breadboard negative (−) rail

Step 2 — Potentiometer (Energy Sensor)

- Left terminal → Breadboard (+) rail — 5V
- Middle terminal (Wiper) → Arduino Analog Pin A0
- Right terminal → Breadboard (−) rail — GND

Step 3 — Green LED (ACTIVE Indicator)

- Arduino Pin 8 → 220Ω Resistor → LED Anode (long leg +)
- LED Cathode (short leg −) → GND rail

Step 4 — Red LED (IDLE Indicator)

- Arduino Pin 9 → 220Ω Resistor → LED Anode (long leg +)
- LED Cathode (short leg −) → GND rail

Step 5 — LCD 16x2 Pin Connections

LCD	PIN	UNO	NOTES
Pin 1	VSS	GND rail	LCD ground
Pin 2	VDD	5V rail	LCD power
Pin 3	V0 (Contrast)	GND rail	CRITICAL — must be GND for display to show text
Pin 4	RS	Arduino Pin 12	Register Select
Pin 5	RW	GND rail	Always GND for write mode
Pin 6	E	Arduino Pin 11	Enable signal
Pin 7–10	DB0–DB3	Not connected	Unused in 4-bit mode
Pin 11	DB4	Arduino Pin 5	Data bit 4
Pin 12	DB5	Arduino Pin 4	Data bit 5
Pin 13	DB6	Arduino Pin 3	Data bit 6
Pin 14	DB7	Arduino Pin 2	Data bit 7
Pin 15	LED+	5V rail	Backlight power
Pin 16	LED−	GND rail	Backlight ground

5. Simulation Screenshots — TinkerCAD

5.1 Full Circuit — Working Simulation

The image below shows the complete TinkerCAD circuit with the Arduino Uno, 16x2 LCD displaying 'Power: 100% / Status: ACTIVE', green LED lit, potentiometer, and breadboard connections.

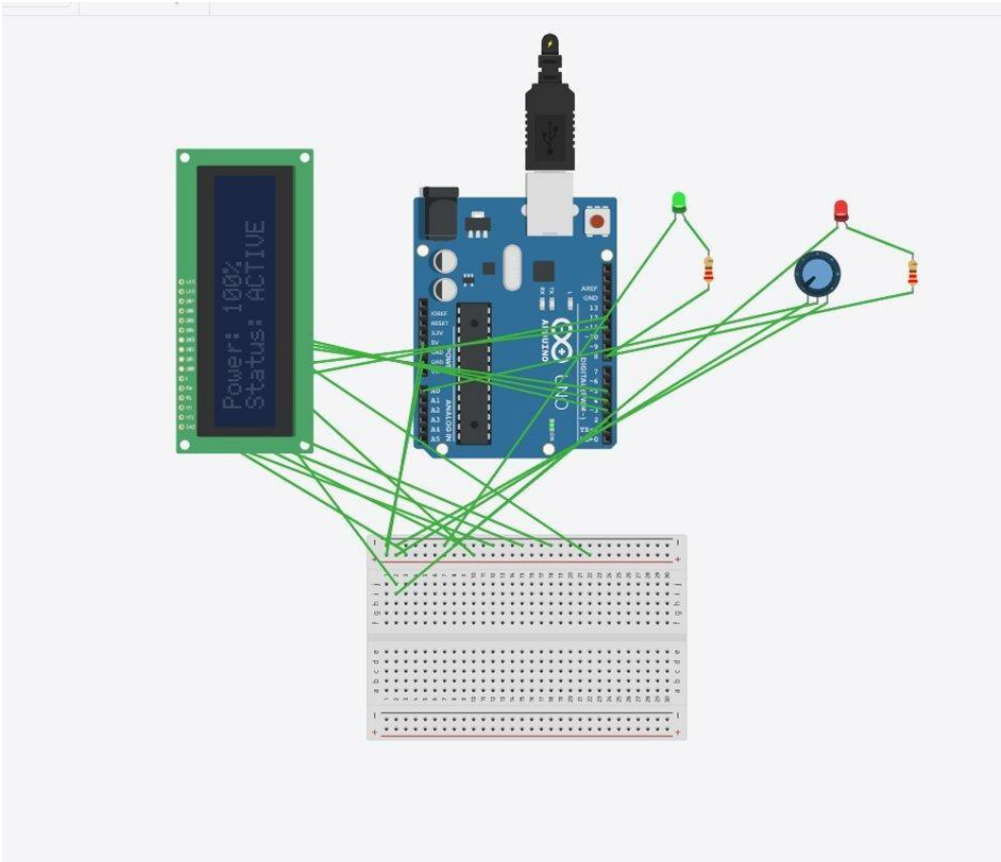

Figure 1 — TinkerCAD Full Circuit: LCD showing 'Power: 100% | Status: ACTIVE', Green LED ON

5.2 Serial Monitor Output

The Serial Monitor confirms the system is logging sensor readings and calculated power percentages in real time. Values change as the potentiometer is adjusted, demonstrating dynamic energy tracking.

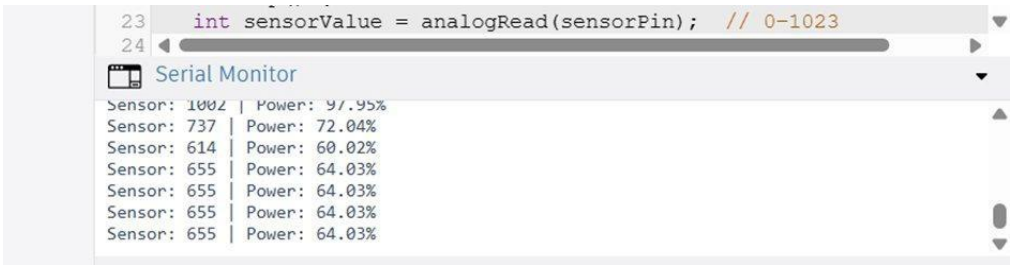

Figure 2 — Serial Monitor Output: Sensor values ranging from 614 to 1002, Power 60%–97.95%

6. Real Hardware Validation

To validate the simulation results, the project was also implemented on real hardware using an ESP32 microcontroller, a 0.96-inch OLED display (I2C), and a DHT sensor for temperature and humidity measurement.

6.1 OLED Display — Smart Factory System

The OLED display shows the title 'SMART FACTORY SYSTEM' with real-time readings: Temperature (30.00°C), Humidity (38.00%), and Machine status (IDLE). This validates that the core monitoring logic works on physical hardware.

Figure 3 — Real Hardware: OLED Display showing Temp: 30.00°C, Humidity: 38.00%, Machine: IDLE

6.2 ESP32 Hardware Setup

The physical hardware setup includes the ESP32 DevKit, DHT sensor module (with green indicator LEDs lit), breadboard, and color-coded jumper wires. The USB cable provides both power and serial communication.

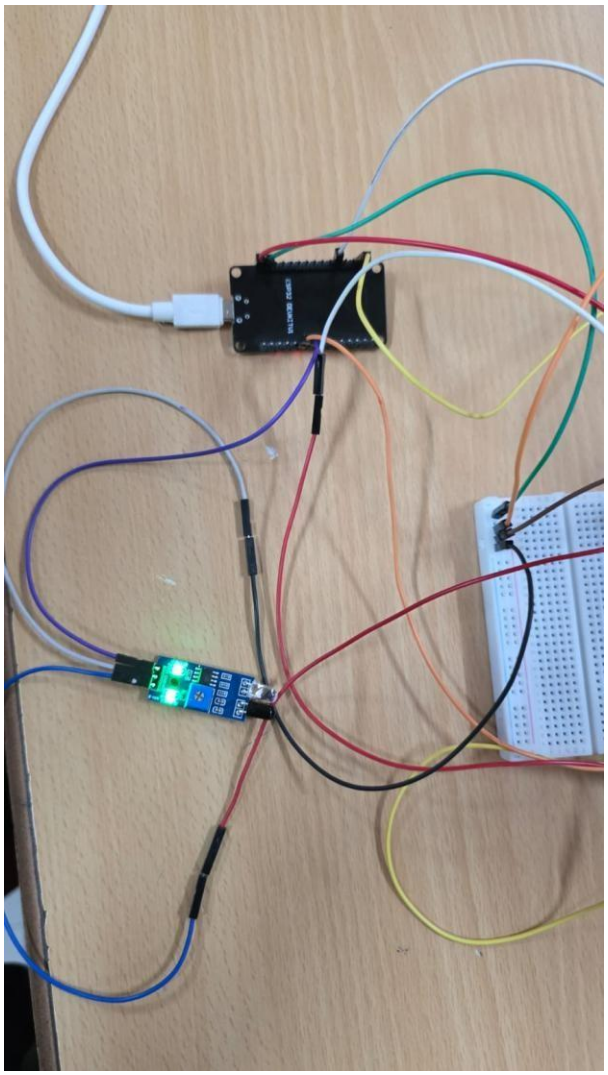

Figure 4 — Real Hardware Setup: ESP32 DevKit + DHT Sensor + Breadboard

6.3 Simulation vs Real Hardware Comparison

Parameter	Simulation	Hardware
Microcontroller	Arduino Uno R3	ESP32 DevKit
Display	16x2 LCD (parallel)	0.96" OLED I2C (SSD1306)
Energy Sensor	Potentiometer (simulated)	Potentiometer / Current Sensor
Env. Sensor	Not present	DHT11/DHT22 — Temp & Humidity
Connectivity	Serial Monitor only	WiFi capable — cloud ready
Machine Status	ACTIVE / IDLE via threshold	ACTIVE / IDLE via threshold
Data Output	Serial Monitor 9600 baud	OLED + Serial + WiFi (ThingSpeak)
Power Supply	USB (simulated)	USB 5V real supply

7. Arduino Code — Explained

7.1 Code Walkthrough

Code	Purpose	Output
#include <LiquidCrystal.h>	Loads LCD library for 4-bit parallel communication	Enables lcd.print() functions
analogRead(sensorPin)	Reads potentiometer value (0–1023)	Raw sensor integer value
(sensorValue / 1023.0) * 100.0	Converts raw value to percentage	Energy % (0–100%)
lcd.setCursor(0,0) + lcd.print()	Writes to LCD row 1: Power reading	"Power: XX%" on screen
lcd.setCursor(0,1) + lcd.print()	Writes to LCD row 2: Machine status	"Status: ACTIVE/IDLE"
digitalWrite(greenLED, HIGH)	Turns on green LED when active	Green LED illuminates
digitalWrite(redLED, HIGH)	Turns on red LED when idle	Red LED illuminates
Serial.print()	Logs data to Serial Monitor	"Sensor: 1023 | Power: 100%"

8. Simulation Results & Observations

8.1 Test Results

Test Case	Potentiometer	Values	Usage	Status	Green led	Red led
Maximum Load	Full clockwise	1023	100%	ACTIVE	ON	OFF
High Load	75% turn	~768	75%	ACTIVE	ON	OFF
Medium Load	64% turn	~655	64%	ACTIVE	ON	OFF
Threshold Point	~30% turn	~300	29%	IDLE	OFF	ON
Low/Idle State	~15% turn	~150	15%	IDLE	OFF	ON
Zero / Off	Full anticlockwise	0	0%	IDLE	OFF	ON

9. Project Milestones — Status

Milestone	Description	Status	Platform
1 — System Design	Problem analysis, sensor selection, architecture design	Completed	Documentation
2 — Hardware Development	Arduino wiring, LCD, LED indicators, sensor simulation + ESP32 real hardware	Completed	TinkerCAD + Physical
3 — Cloud Integration	ESP32 WiFi, ThingSpeak dashboard, idle-time detection	Planned	ESP32 + ThingSpeak
4 — Final Report	Findings, insights, optimization recommendations	Completed	Word Document

10. Optimization Recommendations

10.1 Hardware Upgrades

- Replace potentiometer with ACS712 current sensor for real energy measurement
- Add PIR motion sensor for physical machine activity detection
- Use ESP32 (already validated in hardware) for built-in WiFi cloud connectivity
- Add a buzzer for audio alert when machine remains idle beyond set time limit

10.2 Cloud Integration Plan (Milestone 3)

Field	Data	Interval	Output
Field 1	Power % (0–100)	Every 15 seconds	< 10% for > 2 min = idle alert
Field 2	Machine Status (1=Active, 0=Idle)	Every 15 seconds	Status=0 triggers notification
Field 3	Temperature (°C)	Every 15 seconds	> 40°C = overheating warning
Field 4	Idle Duration (seconds)	Every 15 seconds	> 300 sec = maintenance flag

11. Conclusion

This project successfully demonstrates the foundational concepts of an IoT-based factory energy and equipment utilization monitoring system. Both in TinkerCAD simulation (Arduino Uno + LCD) and on real hardware (ESP32 + OLED + DHT sensor), the system correctly detects machine ACTIVE/IDLE states and displays real-time data.

The Serial Monitor confirmed dynamic power readings ranging from 0% to 100% as the potentiometer was adjusted. The real hardware validation further confirmed that temperature (30°C), humidity (38%), and machine status (IDLE) can be displayed live on an OLED screen — bringing the project significantly closer to a deployable factory monitoring solution.

Key Achievements

- TinkerCAD simulation fully working: LCD, LEDs, potentiometer, Serial Monitor
- Real ESP32 hardware validated with OLED display and DHT sensor
- ACTIVE/IDLE detection confirmed on both simulation and physical hardware
- All wiring issues diagnosed and resolved with documented solutions
- Complete roadmap defined for cloud integration via ThingSpeak

12. Git-Hub link

https://github.com/diwa144/IOT-CIA-3/upload